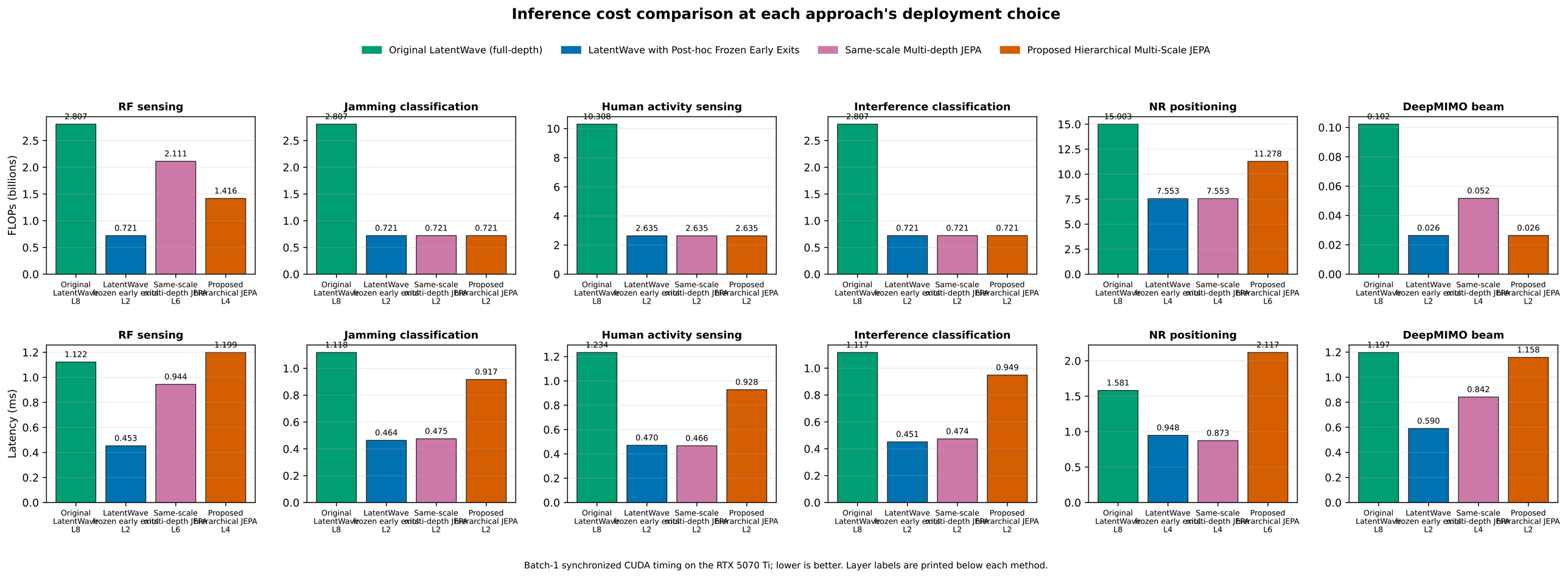
Inference cost comparison at each approach's deployment choice
Original LatentWave (full-depth)
LatentWave with Post-hoc Frozen Early Exits
Same-scale Multi-depth JEPA
Proposed Hierarchical Multi-Scale JEPA
RF sensing
Jamming classification
Human activity sensing
Interference classification
NR positioning
DeepMIMO beam
FLOPs (billions)
Latency (ms)
2.807
0.721
2.111
1.416
10.308
2.635
15.003
7.553
11.278
0.102
0.026
0.052
1.122
0.453
0.944
1.199
1.118
0.464
0.475
0.917
1.234
0.470
0.466
0.928
1.117
0.451
0.474
0.949
1.581
0.948
0.873
2.117
1.197
0.590
0.842
1.158
Original LatentWave L8
LatentWave frozen early exits L2
Same-scale multi-depth JEPA L6
Proposed Hierarchical JEPA L4
Original LatentWave L8
LatentWave frozen early exits L2
Same-scale multi-depth JEPA L2
Proposed Hierarchical JEPA L2
Original LatentWave L8
LatentWave frozen early exits L2
Same-scale multi-depth JEPA L2
Proposed Hierarchical JEPA L2
Original LatentWave L8
LatentWave frozen early exits L2
Same-scale multi-depth JEPA L2
Proposed Hierarchical JEPA L2
Original LatentWave L8
LatentWave frozen early exits L4
Same-scale multi-depth JEPA L4
Proposed Hierarchical JEPA L6
Original LatentWave L8
LatentWave frozen early exits L2
Same-scale multi-depth JEPA L4
Proposed Hierarchical JEPA L2
Batch-1 synchronized CUDA timing on the RTX 5070 Ti; lower is better. Layer labels are printed below each method.